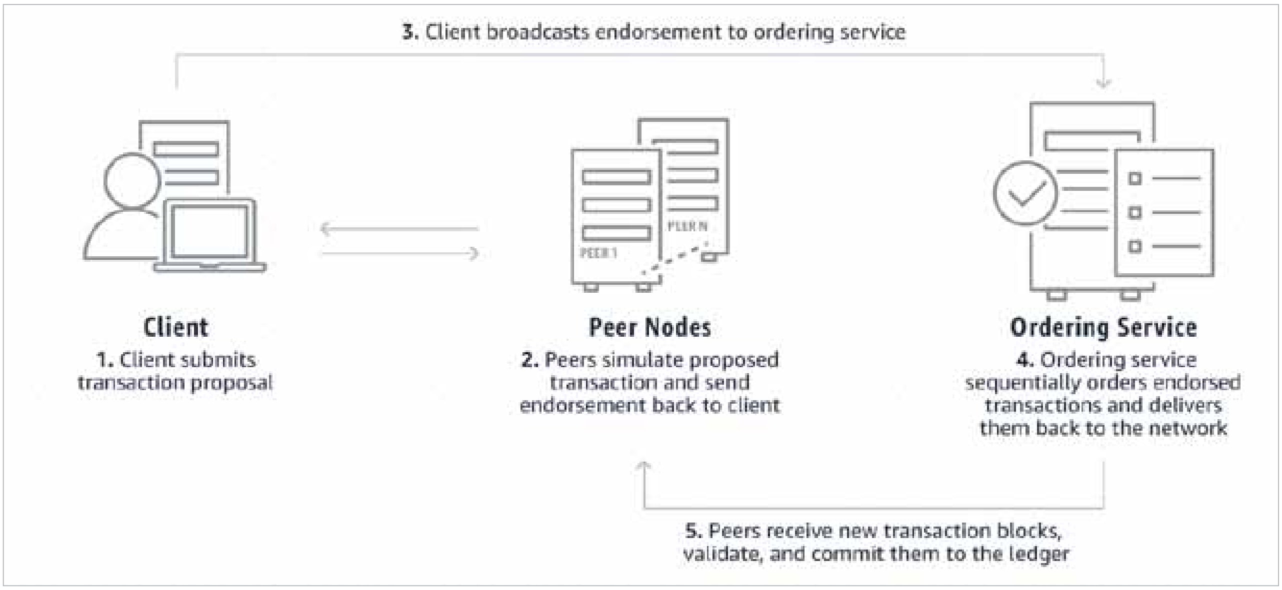

Figure 1: How does Hyperledger Fabric work (Image source: *https://aws.amazon.com/blockchain/what-is-hyperledger-fabric/)*

Smart contracts (Chaincode): Smart contracts in Hyperledger Fabric are implemented using Chaincode, which is a piece of code that defines the rules and logic of transactions. Chaincode is executed on the peers in a secure and isolated environment, ensuring that unauthorised parties cannot tamper with or modify it. This enables automated and trustless execution of transactions, reducing the risk of fraud and errors.

Consensus mechanism: Hyperledger Fabric supports a pluggable consensus mechanism, allowing organisations to choose the consensus algorithm that best suits their needs. This provides flexibility in terms of performance, scalability, and fault tolerance. For example, organisations can choose between the default Kafka-based ordering service or use alternative algorithms like Raft or PBFT for ordering transactions.

Privacy and confidentiality: Hyperledger Fabric ensures the privacy and confidentiality of transactions through channels and private data collections. Channels allow for the creation of private sub-networks within the main network, where only selected participants can access and transact. Private data collections enable parties to store sensitive information off-chain, ensuring it is only shared with authorised parties.

Identity management: Identity management is crucial in blockchain networks to ensure that participants are who they claim to be. Hyperledger Fabric uses the MSP component to manage identities, providing authentication and authorisation services. MSP allows organisations to define their identity management policies, ensuring that only authorised users can access the network.

Scalability: Hyperledger Fabric is designed to be highly scalable, supporting thousands of transactions per second. This is achieved through its modular architecture, which allows for the parallel execution of transactions and using off-chain data storage solutions. Additionally, Hyperledger Fabric supports horizontal scaling, enabling organisations to add more nodes to the network as the demand for transactions increases.

Interoperability: Hyperledger Fabric can interoperate with other blockchain networks and legacy systems through its support for standard protocols and interfaces. For example, it supports the Hyperledger Cactus project, which provides interoperability between different blockchain platforms. This enables organisations to integrate Hyperledger Fabric into existing infrastructure and collaborate with other blockchain networks.

Use cases of Hyperledger Fabric

Here are a few examples of how Hyperledger Fabric is being used in various industries.

1. **Supply chain management:** Hyperledger Fabric is revolutionising supply chain management by providing a transparent and efficient way to track products from their origin to the end consumer. For instance, in the agriculture sector, Hyperledger Fabric is used to trace the journey of produce from farms to store shelves, ensuring its authenticity and quality. This helps in identifying and addressing issues such as contamination or spoilage, leading to safer and more reliable food supply chains.

- Ensures authenticity and quality of products
- Identifies and addresses issues like contamination or spoilage
- Improves supply chain transparency and efficiency

2. **Healthcare:** In the healthcare industry, Hyperledger Fabric is transforming how patient data is managed and shared. By using blockchain technology, healthcare providers can securely store and share patient records, ensuring data privacy and integrity. For example, hospitals